

EnsemHalDet: Robust VLM Hallucination Detection via Ensemble of Internal State Detectors

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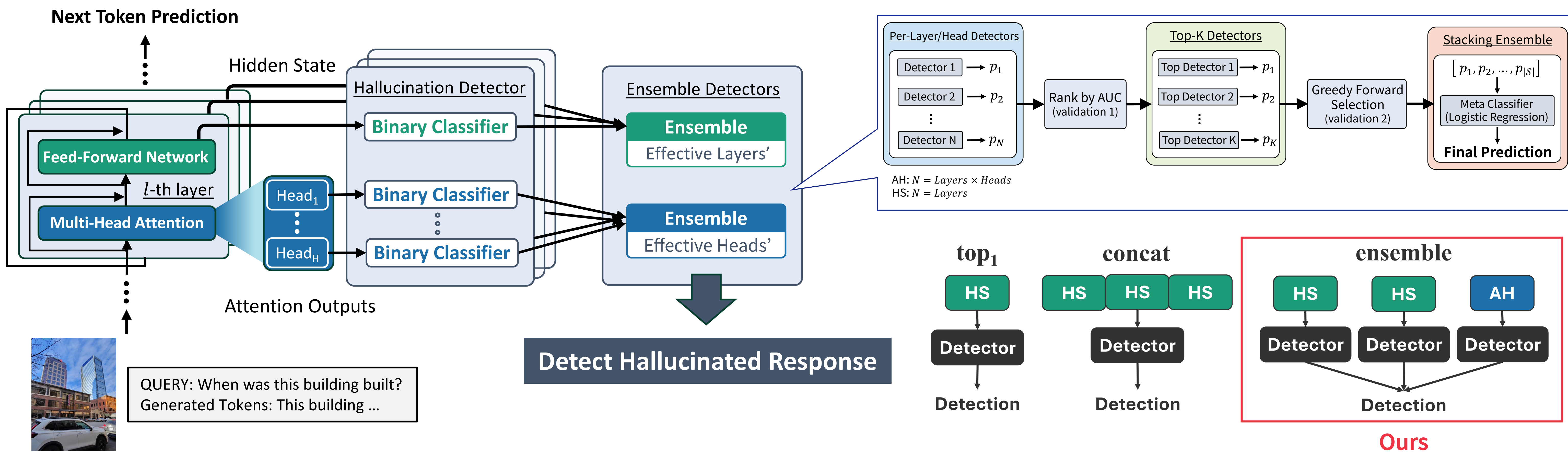
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- Training Independent Hallucination Detectors **on Multiple Internal Representaiton** of VLMs
- Combining Multiple Detectors **via Ensemble Learning** with Optimal Detector Selection
- **High-Accuracy and Efficient Hallucination Detection** Beyond Existing Baselines



EnsemHalDet

Internal Representations

- **AH: Attention head outputs from each head**
- **HS: Hidden states from each layer**

Training unit: each layer/head

- Each detector is independently trained for each internal state.

Dimensionality reduction

- **HS features** are reduced from ~4,000 to 128 dimensions using PCA, because this led to better detection performance.

1. Extract Internal State

- For each generated token $t = 1, \dots, T$, the **final tokens vector** e_t is extracted.
- The feature vector is obtained by **averaging** over all generated tokens:

$$\bar{e} = \frac{1}{T} \sum_{t=1}^T e_t$$

2. Train Detectors

- Training with **logistic regression**:
- $$y = \sigma(w_1^T e + b_1)$$
- y : the response is hallucinated? (1/0)
 - e : internal representation

3. Ensemble Detectors

- Ranking detector (Validation①)
 - Rank all detectors by validation AUC.
 - Select top-performance detectors as candidates, for both **AH** and **HS**
 - **MIX: AH + HS**
- Ensemble selection (Validation②)
 - Apply **greedy forward selection**.
 - Add detectors only when the ensemble AUC improves.
 - **Ensemble detectors with stacking**.
 - Use logistic regression as the meta-classifier for final prediction.

$$p = \sigma(w_2^T \underline{p} + b_2)$$

each detector output

Results

Experimental Setup

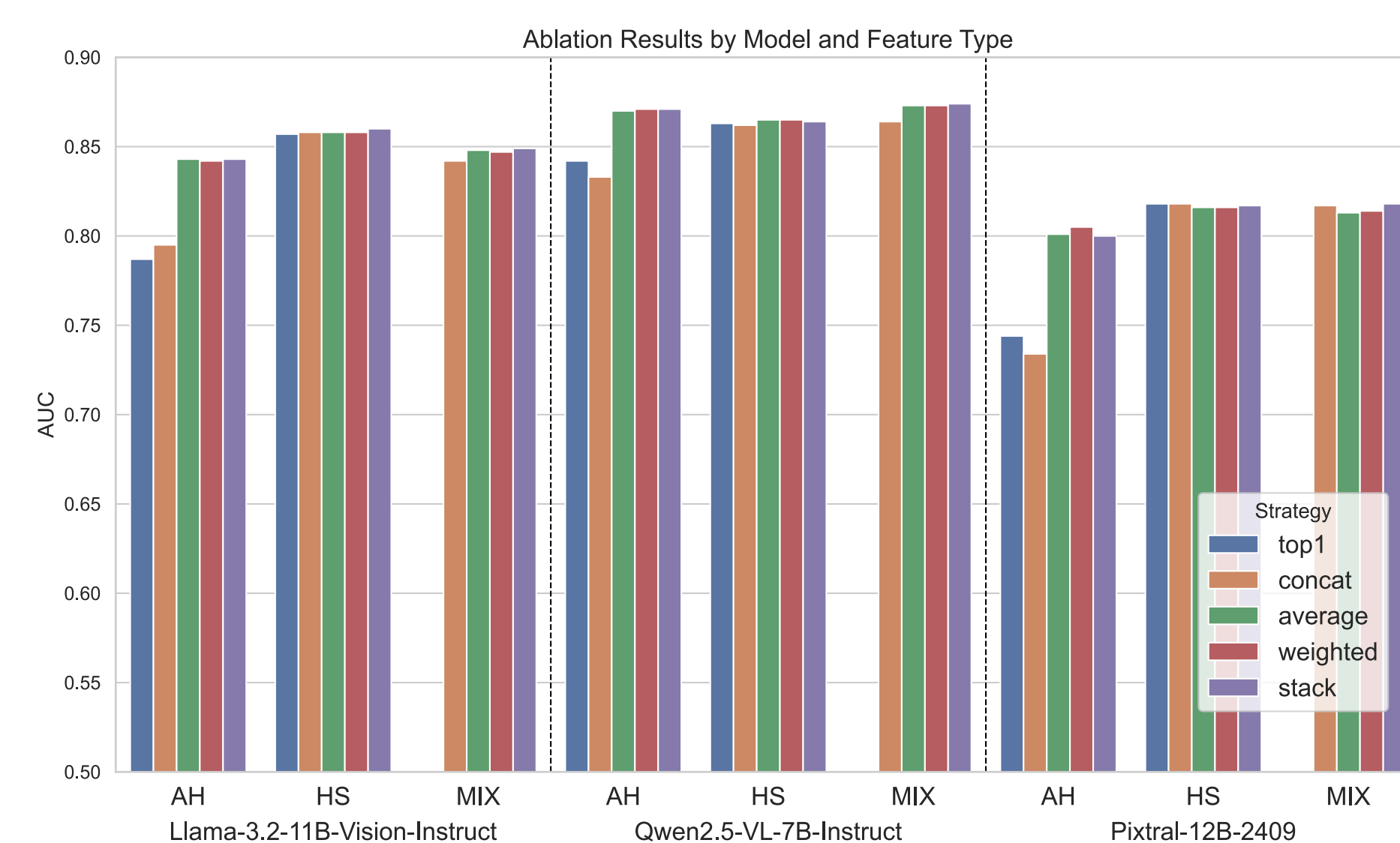
- Task: binary classification with AUC
 - Labels are assigned using LLM-as-a-Judge
 - Compare generated answers with the GT
- Datasets (VQA): CRAG-MM^[1], MMMU-Pro^[2]
- Baselines:
 - probe-based baselines: SAPLMA, MIND, MHAD
 - top1 (single), concat (feature concatenation)
 - ensemble (average, weighted)

EnsemHalDet generally outperforms baselines

Methods	CRAG-MM			MMMU-Pro		
	Llama-3.2-11B	Qwen2.5-VL-7B	Pixtral-12B	Llama-3.2-11B	Qwen2.5-VL-7B	Pixtral-12B
SAPLMA	0.817 ± 0.023	0.797 ± 0.023	0.730 ± 0.045	0.850 ± 0.026	0.615 ± 0.076	0.791 ± 0.015
MIND	0.822 ± 0.011	0.830 ± 0.018	0.750 ± 0.021	0.866 ± 0.039	0.686 ± 0.047	0.801 ± 0.034
MHAD _{att}	0.850 ± 0.019	0.867 ± 0.012	0.784 ± 0.012	0.870 ± 0.033	0.751 ± 0.055	0.822 ± 0.034
MHAD _{hs}	0.838 ± 0.028	0.849 ± 0.011	0.763 ± 0.033	0.889 ± 0.023	0.707 ± 0.056	0.792 ± 0.046
MHAD _m	0.830 ± 0.029	0.855 ± 0.020	0.777 ± 0.038	0.874 ± 0.040	0.742 ± 0.050	0.810 ± 0.042
EnsemHalDet _{att} (ours)	0.843 ± 0.020	0.871 ± 0.021	0.800 ± 0.029	0.857 ± 0.036	0.740 ± 0.036	0.802 ± 0.030
EnsemHalDet _{hs} (ours)	0.860 ± 0.013	0.864 ± 0.012	0.817 ± 0.013	0.860 ± 0.036	0.762 ± 0.021	0.826 ± 0.027
EnsemHalDet _m (ours)	0.849 ± 0.021	0.874 ± 0.019	0.818 ± 0.021	0.854 ± 0.035	0.748 ± 0.037	0.802 ± 0.030

Table 1: Main comparison of hallucination detection methods with AUC. All results are reported as the mean and standard deviation over five runs of feature extraction with different random seeds. Best results are in bold.

Ablation Study



- ✓ Ensembles consistently outperform top1 and concat baselines.
- ✓ HS-based detectors generally perform better than AH-based detectors.
- ✓ AH-based detectors gain more from ensemble than HS-based detectors.
- ✓ MIX (AH+HS) provides only limited additional improvement.
- ✓ Stacking delivers the best overall ensemble performance.

Inference Time

Method	Feature Extraction (s)	Detection (s)	Total (s)	Percentage (%)
Baseline	2.511	—	2.511	100.0
SAPLMA	3.767	0.000060	3.767	150.0
MIND	5.591	0.001465	5.592	222.7
MHAD _{att}	3.095	0.001835	3.097	123.3
MHAD _{hs}	2.952	0.002083	2.954	117.7
MHAD _m	3.182	0.001325	3.183	126.8
EnsemHalDet _{att} (ours)	3.095	0.000023	3.095	123.2
EnsemHalDet _{hs} (ours)	4.618	0.000054	4.618	183.9
EnsemHalDet _m (ours)	5.220	0.000120	5.220	207.9

Table 4: Inference time comparison for hallucination detection. The percentage indicates the total runtime relative to the baseline generation time (100%). We report GPU-based inference results for SAPLMA, MIND, and MHAD.

- ✓ Feature extraction dominates the total inference time.
- ✓ EnsemHalDet achieves much faster detection than MLP-based baselines.

Discussion

- ✓ Theoretical explanation for why detector ensembles outperform a single detector
- ✓ Robustness under domain transfer
- ✓ Bottleneck caused by the time cost of internal representation extraction

Reference

- [1]. Wang et al., **CRAG-MM: Multi-modal Multi-turn Comprehensive RAG Benchmark**, 2025, arXiv:2510.26160
- [2]. Yue et al., **MMMU-Pro: A More Robust Multi-discipline Multimodal Understanding Benchmark**, 2025, ACL2025